Home Lab 2: The Powder Workflow

Do this with a parent. Goggles on. Never taste anything. Clean up after. See the Home Lab Safety Agreement.

Goal

Practice running the same tests in the same order on four known powders, so it becomes automatic.

You need

From the kit: **baking soda, corn starch, table salt, sugar**; iodine dropper; cabbage indicator. From home: white vinegar, water, a muffin tin or 4 cups, toothpicks, paper towels.

Steps

For **each** of the four powders, in this order:

1. **Look** — write down what it looks like (crystal? fine powder?).
2. **Feel** — rub a pinch between your fingers. Gritty? Silky? Soft?
3. **Water** — stir a pinch into a little water. Does it fully dissolve, stay cloudy, or make "oobleck"?
4. **pH** — add 2 drops of cabbage indicator to the water mix. Color? Acid, neutral, or base?
5. **Acid** — put a fresh pinch on a paper towel, add a few drops of vinegar. Fizz or no fizz?
6. **Iodine** — put a fresh **dry** pinch on the paper towel, add **one** drop of iodine. Blue-black, or stays orange-brown?

Do all four. Then do it a second time, faster, and see if you can do it from memory without checking the steps.

Results

Powder	Look	Feel	In water	pH (cabbage)	+ Vinegar	+ Iodine
Baking soda						
Corn starch						
Table salt						
Sugar						

Follow-up

1. Which powder fizzed with vinegar? What gas made the fizz?
2. Which powder turned iodine blue-black? What does that tell you is in it?

3. Which powder(s) had a basic pH?
4. Give the chemical formula for each of the four powders.

Coach notes

- Baking soda: soft, dissolves, basic (blue-green), **fizzes**, no iodine change. $NaHCO_3$.
- Corn starch: silky/squeaky, "oobleck," neutral, no fizz, **blue-black iodine**. $(C_6H_{10}O_5)_n$.
- Table salt: cubic, dissolves, neutral, no fizz, no iodine change. NaCl.
- Sugar: glassy grains, dissolves, neutral, no fizz, no iodine change. $C_{12}H_{22}O_{11}$.
- Follow-up: (1) baking soda; carbon dioxide. (2) corn starch; starch. (3) baking soda. (4) as above.